

# Capstone controlled experiments

Project: Image-pattern Network Investigation

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Capstone hypothesis and baseline

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Efficiency and resource measurement

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Transfer learning

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Trace both runs for Capstone controlled experiments. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Capstone controlled experiments. Proposed explanation: “More runs mean more validation examples”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Run a new comparison independently

For training (1,3),(2,6) and validation (3,9),(4,12), start  $a = 1$ . Compare rates 0.1 and 0.2 for two updates. Gradient is  $5*(a-3)$ . Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. W1 Name the controls

List six settings held fixed and the single intended changed factor.

### My reasoning and evidence

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## 6. W2 Trace both runs

Compute both updates for rates 0.1 and 0.2 using gradient  $5*(a-2)$ .

### My reasoning and evidence

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## 7. W3 Select without extra evidence

Recompute validation MSE for the final weights and apply the written selection rule. How many distinct validation rows were used?

### My reasoning and evidence

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## 8. W4 Run a new comparison independently

For training (1,3),(2,6) and validation (3,9),(4,12), start  $a = 1$ . Compare rates 0.1 and 0.2 for two updates. Gradient is  $5*(a-3)$ . Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

### My reasoning and evidence

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## 9. W5 Audit a mixed comparison

Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different training rows. Can the difference be attributed to rate alone? Propose a repair.

### My reasoning and evidence

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## 10. Project checkpoint and reflection

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final evidence pack: For training (1,3),(2,6) and validation (3,9),(4,12), start  $a = 1$ . Compare rates 0.1 and 0.2 for two updates. Gradient is  $5 \cdot (a-3)$ . Compute final weights and validation MSE, then select lower MSE; ties go to 0.1. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

### My project change, supporting evidence and remaining limit

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## 11. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 12. Pictorial case practice: Capstone controlled experiments

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1]. Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

AI ACADEMY / AI-5/34

### Controlled toy-pattern comparison

correct labels out of 6 shared patterns

A



B



Bar length shows the supplied value; the marked baseline is zero.

## Capstone controlled experiments

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Identify the intervention

B adds the centre-feature connection; all other declared settings stay fixed.



### 2. Count outcomes

A is 4/6 correct; B is 5/6.



### 3. Inspect paired cases

Only case three changes correctness; case five fails for both.

### CHANGE ONE THING

Change B's training set as well as the feature connection.

Explain the changed result using the matching lesson.

Compute the two correct counts and identify the changed case. Why would changing B's training data weaken the conclusion about its added connection?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

## 13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

### Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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## 14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only case three changes correctness; case five fails for both.

1. Inspect paired cases
2. Identify the intervention
3. Count outcomes

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Inspect paired cases Only case three changes correctness; case five fails for both. Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

### Practice 2: explain the deciding evidence

Changed-case practice: Change B's training set as well as the feature connection. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The experiment no longer isolates the connection change.

### **My selected conclusion, deciding fact and explanation**

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The experiment no longer isolates the connection change. A training-data difference supplies an alternative explanation for the observed result.

### **Repair a missing building block**

Review Capstone hypothesis and baseline (ai-5/33). Goal: Write a testable capstone hypothesis and a credible baseline. Short practice: Name the intended change, the baseline and the measured outcome. Why must both versions avoid training on the eight reserved labels?. Return to this week and explain what you changed.

Review Efficiency and resource measurement (ai-5/31). Goal: Calculate throughput and amortised time per example from repeated timings. Short practice: Why is 10ms/case not necessarily a single-request latency? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted output head. Short practice: Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

### **Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

### **What I repaired, the evidence I kept and the next test**

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